



Reading: Exemplar: Software packaging and file archiving on Linux

Introduction

This lab focused on installing and removing software in Linux, and working with zipped files. You installed a text editor called Sublime Text, and extracted/un-extracted .tar files. You also used **apt** to install/uninstall programs.

This exemplar is a walkthrough of the previous Qwiklab activity, including detailed instructions and solutions. You may use this exemplar if you were unable to complete the lab and/or you need extra guidance in competing lab tasks. You may also refer to this exemplar to prepare for the graded quiz in this module.

Start the lab

You'll need to start the lab before you can access the materials. To do this, click the green **Start Lab** button at the top of the screen.

A green rectangular button with rounded corners and the text "Start Lab" in white.

After you click the **Start Lab** button, you will see a shell, where you will be performing further steps in the lab. You should have a shell like this:

```
student@864a6934570a:~$
```

Installing Sublime Text

First, use **dpkg** to install a text editor, called **Sublime Text**. A .deb file is located at **/home/qwiklab/downloads/sublime-text_build-3211_amd64.deb**, which you can install using this command:

```
sudo dpkg -i /home/qwiklab/downloads/sublime-text_build-3211_amd64.deb
```

OUTPUT:

```
Selecting previously unselected package sublime-text.  
(Reading database ... 36788 files and directories currently installed.)  
Preparing to unpack .../sublime-text_build-3211_amd64.deb ...  
Unpacking sublime-text (3211) ...  
dpkg: dependency problems prevent configuration of sublime-text:  
sublime-text depends on libgtk-3-0; however:  
Package libgtk-3-0 is not installed.  
  
dpkg: error processing package sublime-text (--install):  
dependency problems - leaving unconfigured  
Processing triggers for mime-support (3.62) ...  
Processing triggers for hicolor-icon-theme (0.17-2) ...  
Errors were encountered while processing:  
sublime-text
```

You'll notice some errors after this command, but that's okay! The Sublime Text package has some dependencies that aren't installed on your machine yet, and **dpkg** is just warning you that they need to be installed. You can remedy this by using **apt** to fix the missing dependencies, using this command:

```
sudo apt install -f
```

You'll be prompted to confirm your decision to continue with the operation by typing **Y** (for "Yes"), midway through the process.

Now Sublime Text is successfully installed, which you can verify using **dpkg**:

```
dpkg -s sublime-text
```

OUTPUT:

```
Package: sublime-text  
Status: install ok installed  
Priority: optional  
Section: editors  
Installed-Size: 34033  
Maintainer: Sublime HQ Pty Ltd  
Architecture: amd64  
Version: 3211
```

Extracting an archive

Next, you will extract a .tar archive. The archive **extract_me.tar** is located in **/home/qwiklab/downloads/**. Move to that directory using this command:

```
cd /home/qwiklab/downloads
```

You can use the Linux tar command to extract it using this command:

```
sudo tar -xvf extract_me.tar
```

The contents of the archive (the file named **great_job**) are then extracted:

OUTPUT:

```
home/qwiklab/extract_me/  
home/qwiklab/extract_me/great_job
```

Archive files

First, move back to the original directory:

```
cd ~
```

You can also use the **tar** command to do the reverse operation, creating an archive. There are three files in your **/home/qwiklab/documents** folder named **Earth**, **Mercury**, and **Venus**. Use tar to archive them into the file **Planets.tar** using this command (one line):

```
tar -cvf Planets.tar --absolute-names /home/qwiklab/documents/Earth \  
/home/qwiklab/documents/Mercury /home/qwiklab/documents/Venus
```

Planets.tar will then be added to your current directory, and will contain the three planet files:

1. **/home/qwiklab/documents/Earth**
2. **/home/qwiklab/documents/Mercury**
3. **/home/qwiklab/documents/Venus**

Installing 7-Zip

You can also install programs on Linux, using **apt** too, which handles dependencies for you to simplify the installation process. You can install the program 7-Zip, using **apt**, with a simple one-line command:

```
sudo apt install p7zip-full
```

You'll be prompted to confirm your decision midway through the installation process. Confirm this by typing **Y** (for "Yes"). Then, 7-Zip will be installed:

OUTPUT:

```
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  p7zip
Suggested packages:
  p7zip-rar
The following NEW packages will be installed:
  p7zip p7zip-full
0 upgraded, 2 newly installed, 0 to remove and 21 not upgraded.
Need to get 1540 kB of archives.
After this operation, 5780 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://deb.debian.org/debian buster/main amd64 p7zip amd64 16.02+dfsg-6 [376 kB]
Get:2 http://deb.debian.org/debian buster/main amd64 p7zip-full amd64 16.02+dfsg-6 [1164 kB]
Fetched 1540 kB in 0s (8625 kB/s)
debconf: delaying package configuration, since apt-utils is not installed
Selecting previously unselected package p7zip.
(Reading database ... 48397 files and directories currently installed.)
Preparing to unpack .../p7zip_16.02+dfsg-6_amd64.deb ...
Unpacking p7zip (16.02+dfsg-6) ...
Selecting previously unselected package p7zip-full.
Preparing to unpack .../p7zip-full_16.02+dfsg-6_amd64.deb ...
Unpacking p7zip-full (16.02+dfsg-6) ...
Setting up p7zip (16.02+dfsg-6) ...
Setting up p7zip-full (16.02+dfsg-6) ...
```

You can verify the installation using **dpkg**, with this command:

```
dpkg -s p7zip-full
```

OUTPUT:

```
Package: p7zip-full
Status: install ok installed
Priority: optional
Section: utils
Installed-Size: 4659
Maintainer: Robert Luberda<robert@debian.org>
Architecture: amd64
Source: p7zip
Version: 16.02+dfsg-6
```

Uninstalling GIMP

Uninstallation can also be handled by **apt** by using **remove** instead of **install** as the argument. GIMP, an image-editing program, is already installed on your machine. Uninstall it now, using **apt** with this command:

```
sudo apt remove gimp
```

You'll be prompted to confirm your decision midway through the uninstallation process. Confirm this by typing **Y** (for "Yes"). Then, GIMP will be uninstalled:

OUTPUT:

```
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following packages were automatically installed and are no longer required:
 fonts-droid-fallback fonts- noto-mono ghostscript gimp-data gsfonts
 i965-va-driver intel-media-va-driver libaa1 libamd2 libaom0 libasound2
 libasound2-data libasynccns0 libavcodec58 libavformat58 libavutil56
 libbabl-0.1-0 libblas3 libcaca0 libcamd2 libccolamd2 libcholmod3
 libchromaprint1 libcodec2-0.8.1 libcolamd2 libcrystalhd3
 libcupsfilters1 libcupsimage2 libde265-0 libexiv2-14 libflac8
 libgegl-0.4-0 libgegl-common libgexiv2-2 libgfortran5 libgimp2.0
 libgme0 libgs9 libgs9-common libgsm1 libheif1 libigdgmm5 libijs-0.35
 libilmbase23 libjbig2dec0 liblapack3 libmetis5 libmng1 libmp3lame0
 libmpeg123-0 libmypaint-1.3-0 libmypaint-common libnuma1 libogg0
 libopenexr23 libopenjp2-7 libopenmpt0 libopus0 libpaper-utils libpaper1
 libpoppler-glib8 libpoppler82 libpulse0 libraw19 libSDL1.2debian
 libshine3 libslang2 libsnappy1v5 libsndfile1 libsoxr0 libspeex1
```

```
libssh-gcrypt-4 libsuitesparseconfig5 libswresample3 libswscale5  
libtheora0 libtwolame0 libumfpack5 libva-drm2 libva-x11-2 libva2  
libvdpau-va-gl1 libvdpau1 libvorbis0a libvorbisenc2 libvorbisfile3  
libvpx5 libwavpack1 libwebpdemux2 libwebpmux3 libwmf0.2-7 libx264-155  
libx265-165 libxcb-xfixes0 libxvidcore4 libzvbi-common libzvbi0  
mesa-va-drivers mesa-vdpau-drivers poppler-data va-driver-all  
vdpau-driver-all
```

Use 'sudo apt autoremove' to remove them.

The following packages will be REMOVED:

```
gimp
```

0 upgraded, 0 newly installed, 1 to remove and 21 not upgraded.

After this operation, 19.5 MB disk space will be freed.

Do you want to continue? [Y/n] y

(Reading database ... 48496 files and directories currently installed.)

Removing gimp (2.10.8-2) ...

Processing triggers for mime-support (3.62) ...

Processing triggers for desktop-file-utils (0.23-4) ...

You can verify the uninstall using **dpkg** (like before) with this command:

```
dpkg -s gimp
```

OUTPUT:

```
dpkg-query: package 'gimp' is not installed and no information is available  
Use dpkg --info (= dpkg-deb --info) to examine archive files.
```

Conclusion

Congratulations! You've successfully installed and uninstalled programs on Linux, and archived and unarchived .tar files.
